more tokens over the required optimal token length would not be helpful. We also computed Pearson correlation in Figure 10, which shows the same trend.

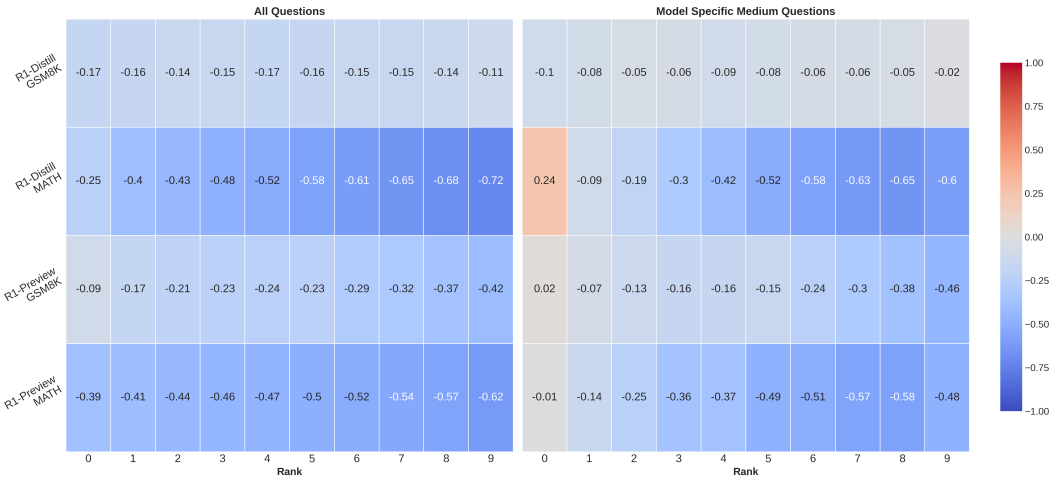

Figure 9: Spearman correlation of answering length and correctness.

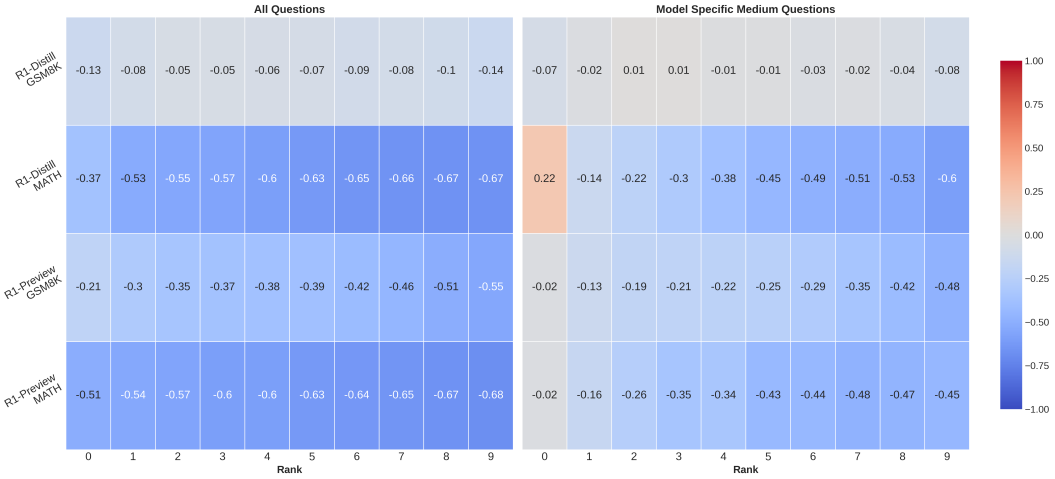

Figure 10: Pearson correlation of answering length and correctness.

C Statistics for Each Model Specific Difficulty Level

Table 4: The number of questions in each model-specific difficulty level.

Dataset	Model	Easy	Medium	Hard	Total
GSM8K	R1-Preview	4809	2377	287	**7473**
	R1-Distill	2118	4967	388	
MATH	R1-Preview	5354	1424	722	**7500**
	R1-Distill	3489	3122	889	

The number of questions in each category and their average response token length are illustrated in Table 4 and Figure 11 respectively.

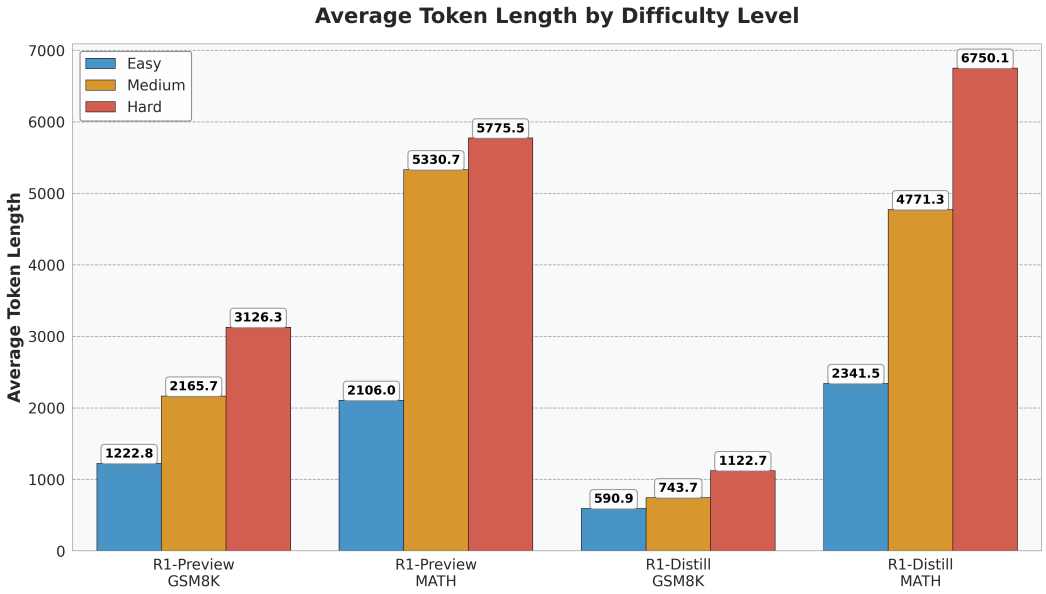

Figure 11: Average token length of responses in model-dataset specific difficulty level.

D Pair-wise Heatmap of Accuracy, Token Length and Perplexity

Length Versus Accuracy Figure 12 shows the heatmap of generation length and accuracy. For easier questions (i.e., questions with high accuracy), the generation lengths tend to cluster around shorter responses. However, for extremely challenging questions (i.e., questions with accuracy 0), the generation lengths are widely dispersed across the entire range, rather than clustering around the longest response length possible.

Perplexity Versus Accuracy Figure 13 shows the heatmap of perplexity and accuracy. Unlike generation length, perplexity doesn't vary much between easy and hard questions overall, most of the perplexity fall into the same range for questions with various accuracy level. However, we observe for more challenging questions (i.e., those with low accuracy), there are more prominent high-perplexity outliers. In contrast, easier questions exhibit a tighter perplexity distribution with fewer extreme values.

Length Versus Perplexity Figure 14 shows the heatmap of generation length versus perplexity. The majority of questions cluster in the region with both low perplexity and short generation length.

E Length Reduction using SimPO

Parameters In Table 5, we list the parameters and compute we use when using SimPO to reduce the generation length.

Additional Results In Figure 15, we show the accuracy and token length trade-offs during SimPO training steps. In Figure 16, we show token length reduction rate for correct answer and incorrect answer during SimPO training steps. For GSM8K dataset, the length reduction on the incorrect responses are significantly high than that of correct responses. While for MATH dataset, the length reduction rate is similar for both correct and incorrect responses.

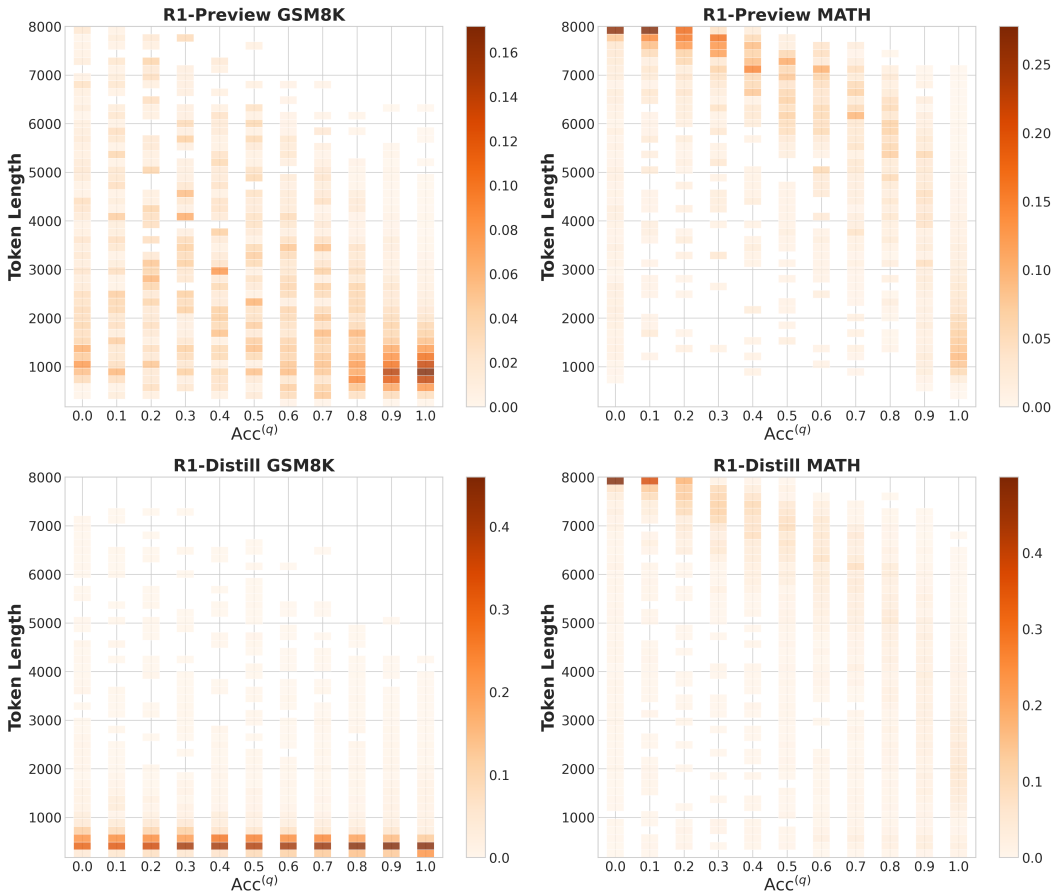

Figure 12: Heatmap of token length for questions categorized by accuracy. Density is normalized among the questions with that accuracy.

Table 5: Training Hyperparameters using SimPO.

Parameter	**Value**
Learning Rate	5e-7
Batch Size (per device)	1
Gradient Accumulation Steps	4
SimPO Parameter β	2
SimPO Parameter γ	1
GPU	2 A100 80G